

Ember

The long life of a low-mass star

Stellar evolution toward an ultra-cold helium remnant

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Current result. Ember has evolved a star of $0.1M_{\odot}$ through 3.40×10^{12} yr from a static initial model at the beginning of hydrogen burning. The star remains fully convective: convection mixes material through almost its entire mass. Its hydrogen mass fraction is 0.2034 and its effective surface temperature is 3202 K. It has not exhausted hydrogen or become a helium white dwarf. The aim is to calculate that transition and the subsequent cooling to surface temperatures of 1000, 500 and 100 K.

The calculations below compare Ember with an independently run Fortran 77 stellar-evolution program, here called F77. They show substantial differences in luminosity and hydrogen-consumption time. Understanding these differences is part of the work; neither agreement with an older result nor greater computational expense establishes physical accuracy. Throughout this paper, $1\text{Tyr} = 10^{12}$ yr. Reported values have at most four significant figures; the files used to calculate the figures retain their available precision.

1 The star and its evolution so far

The initial hydrogen mass fraction is $X_{\text{H}} = 0.7$, the initial ${}^3\text{He}$ fraction is zero, and the metal mass fraction is $Z = 0.02$. The metals follow the relative elemental abundances of Grevesse and Sauval’s GS98 mixture. Here “metals” means elements heavier than helium. The remaining initial mass is ${}^4\text{He}$. The reference holds baryonic mass fixed and omits rotation, magnetism, mass loss and external heating. Its age excludes formation and pre-main-sequence contraction.

Quantity	3.30Tyr	3.40Tyr
Radius ($R_{\odot}$)	0.1470	0.1459
Photon luminosity ($L_{\odot}$)	0.001978	0.002016
Effective surface temperature (K)	3175	3202
Central temperature (MK)	7.768	8.031
Central density (g cm^{-3})	252.3	258.3
Central hydrogen mass fraction	0.2223	0.2034
Central ${}^3\text{He}$ mass fraction	0.001358	0.0009703
Remaining hydrogen mass ($M_{\odot}$)	0.02223	0.02034
Convective mass fraction	1	1

The model uses 512 mass points. The calculation to 3.30Tyr accepted 1822 time steps and repeated three steps at smaller sizes. Continuing the saved model to 3.40Tyr required another 38 accepted steps and no further repeats. Both parts used the same executable and physical inputs. Their common state was checked before combining the histories into 1861 distinct saved models. The decline in ${}^3\text{He}$ and the recent decrease in radius occur while the star is still burning hydrogen, not during white-dwarf cooling.

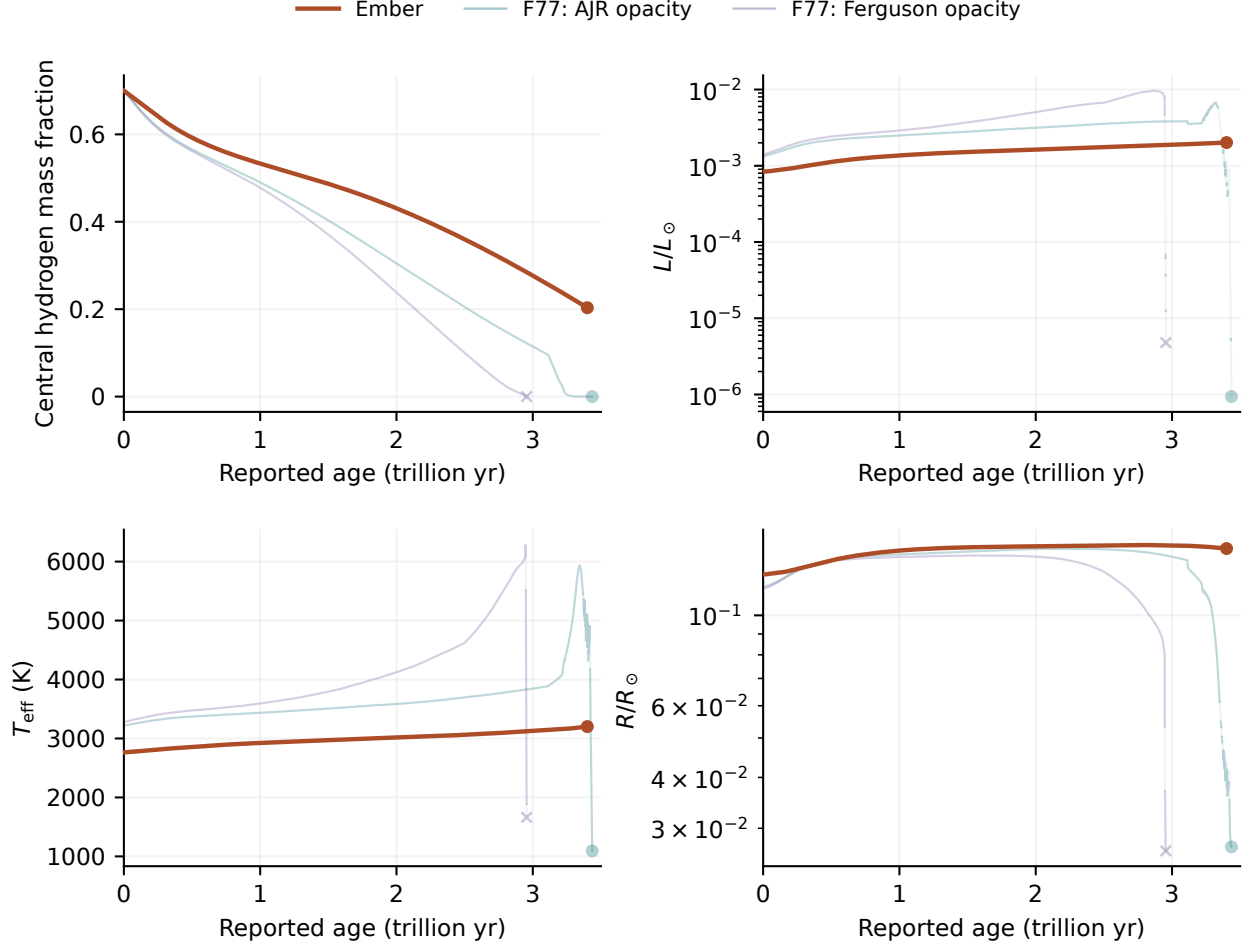


Figure 1: Ember and two F77 calculations of a $0.1M_{\odot}$ star. F77 uses either AJR or Ferguson low-temperature opacity; its other input choices are held fixed. Ember ends at 3.40Tyr while still burning hydrogen. The F77 AJR curve continues through its programmed cooling stop; crosses mark the last accepted Ferguson model before its later failure to converge. These are different physical models. The clocks are shown as calculated: F77 includes contraction before hydrogen burning, whereas Ember begins from a static hydrogen-burning model. Only F77’s first billion years are omitted from this overview. Luminosity and radius use logarithmic scales and the same nominal solar units for all curves. The rapid F77 transitions near the ends of the tracks are resolved in the saved data, although compressed on this time axis. Thin, translucent F77 lines keep attention on Ember. Individual steps changing a surface quantity by more than 5%, or an abundance by more than 0.01, are drawn more faintly; no accepted data are removed or smoothed.

2 What differs from the F77 calculation?

Ember is a C++23 development of the stellar-evolution methods reconstructed in [Heney/F77](#). The long-term evolutionary picture of Laughlin, Bodenheimer and Adams [1] motivates the work; its lifetime is not imposed on either of the new calculations shown here. F77 was run from the retained source with explicitly recorded input files, 151 initial mass points and a mixing-length parameter of 1.36. These runs do not recover an unarchived executable or establish the precise result of the original 1997 calculation.

Age alone can give a misleading comparison because the two codes consume hydrogen at different rates. Figure 2 instead compares stars at the same central hydrogen abundance. Near Ember’s present $X_{\text{H}} = 0.2034$, the F77 AJR star is about 76% more luminous. In common solar units its luminosity is approximately $0.003554L_{\odot}$, compared with Ember’s $0.002016L_{\odot}$. Their surface temperatures are 3698 and 3202 K; their radii and central temperatures differ by less than 0.5%. F77 reaches this abundance near 2.507Tyr, compared with Ember’s 3.40Tyr, and retains about 56% more ^3He .

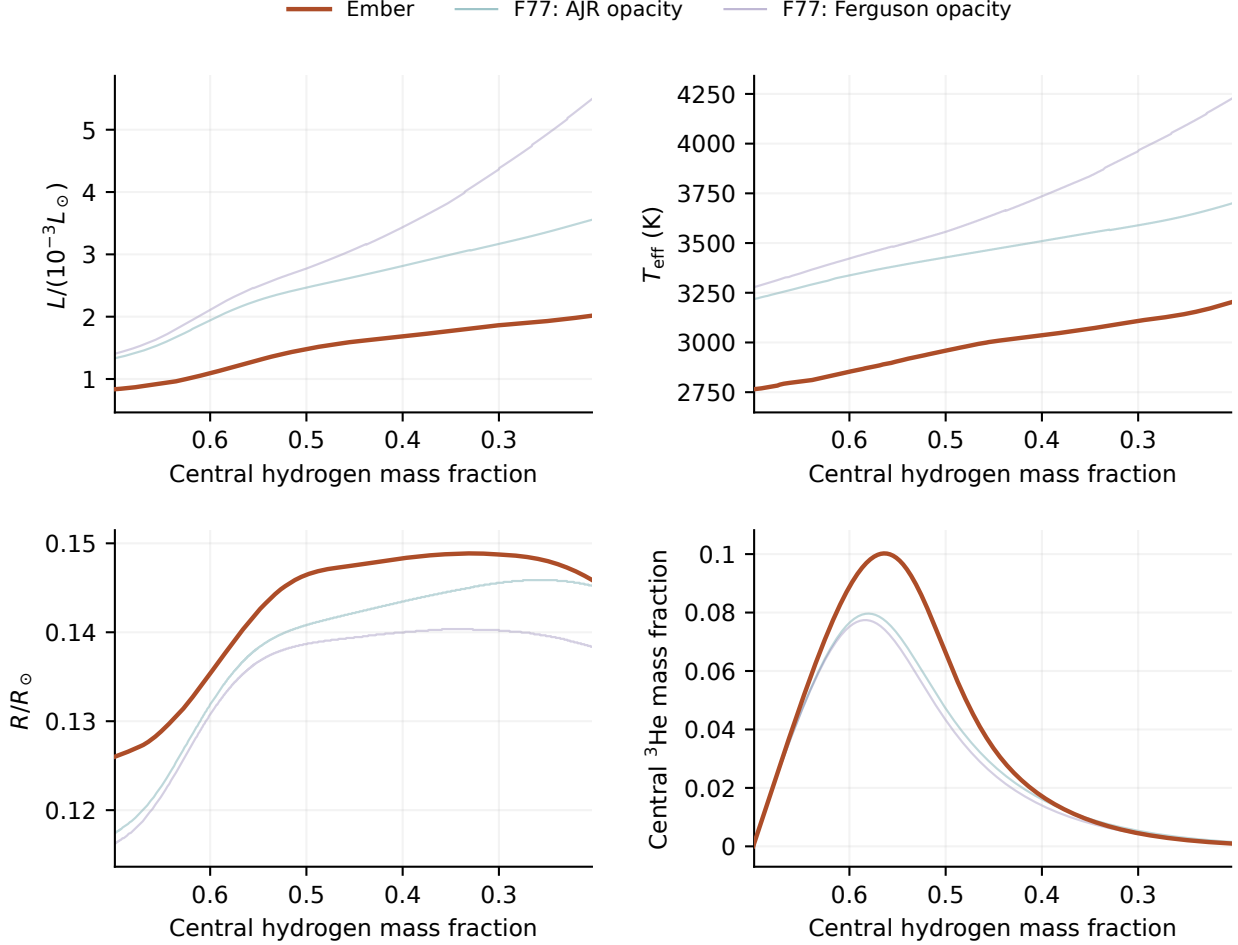


Figure 2: Comparison at the same stage of hydrogen consumption. Hydrogen abundance decreases from left to right, over the interval reached by all three calculations. Luminosity, surface temperature, radius and central ^3He abundance are plotted directly against the central hydrogen fraction; no stellar age is shifted and no track is extrapolated. Values above $X_{\text{H}} = 0.699$ are omitted to exclude the initial F77 contraction and very early adjustment. Each line connects accepted models. F77 surface quantities were printed to four significant figures. Its luminosity and radius have been converted to Ember’s nominal solar units. The differences measure the combined effect of the codes’ physical and numerical choices, not the isolated error of any one ingredient. The faint-line treatment is the same as in Figure 1.

Changing only F77’s low-temperature opacity choice from AJR to Ferguson increases luminosity by about 54.5% at this hydrogen abundance. In a separate calculation, the two codes’ pp-chain routines were evaluated at identical central temperature, density, hydrogen and ^3He abundance. Their heating rates differ by roughly 9%. This is much smaller than the surface-luminosity difference between the independently evolved stars.

These results make the atmosphere and opacity, together with the resulting ${}^3\text{He}$ history, leading possibilities to investigate. They do not uniquely separate those effects from the equation of state, nuclear screening or the way composition and structure are solved together. Baraffe et al. [3] discuss how grey atmospheres can overestimate surface temperature and luminosity in cool low-mass stars. That supports the direction of a possible effect, not its size here. A decisive comparison will require relaxing otherwise equivalent stars while changing one physical ingredient at a time. F77 already includes CNO burning, conduction and an adjustable mass mesh; Ember does not simply add physics to a smaller F77 model.

2.1 When does the center cease to be convective?

At 3.40Tyr, Ember’s smallest ratio of the temperature gradient required by radiation and conduction to the adiabatic gradient is 1.336. It occurs near 21.1% of the enclosed mass; the central ratio is 3.090. At 3.30Tyr the corresponding ratios were 1.588 and 3.707. For homogeneous material, a ratio below unity permits stable transport without convection. The decreasing ratios suggest an approaching change, but do not yet establish a radiative core or its formation time. These estimates use values at mass points; Ember determines mixing between adjacent points during evolution.

F77 was given an additional print statement to record its central convection flag. Removing these extra lines reproduces the unmodified program’s output exactly for the full AJR run and a shorter, 6000-model Ferguson comparison. The first accepted models with a nonconvective central flag are:

F77 opacity	Age (Tyr)	Central hydrogen mass fraction
AJR	3.224	0.01756
Ferguson	2.946	1.064×10^{-5}

F77 changes its convection flag only after a finite margin beyond marginal stability. For AJR, the gradient first fell below unity one accepted model before the flag changed. The exact bounding models are retained in the data; the rounded ages above cannot resolve that one-step difference. The extended Ferguson run later failed to converge during cooling near 2.955Tyr. Its accepted earlier models remain useful, but that run did not finish cooling.

3 Physical ingredients in Ember

Component	Treatment and present limits
Equation of state	FreeEOS 3.0 supplies pressure, internal energy and their responses to temperature, density and composition. Interpolation is based on a shared Helmholtz free energy to keep these responses consistent. The selected 64 composition tables include zero hydrogen, within a limited temperature and density range. They do not yet describe the cold, dense remnant.
Opacity	AESOPUS 2.1 provides low-temperature gas opacity and LANL TOPS provides hotter and denser conditions. The two are joined only where their available ranges permit it. Hydrogen-poor compositions have been refined and checked against additional calculations.

Component	Treatment and present limits
Hydrogen burning	A reduced ppI/ppII network follows ${}^3\text{He}$ out of equilibrium. It uses Solar Fusion II reaction data and finite-degeneracy Salpeter–Van Horn screening, which accounts for the plasma’s effect on charged reactants. Atomic mass differences determine heat and nuclear-neutrino energy. Pep, hep, ppIII and CNO reactions are absent; metals do not react.
Heat and composition transport	Mixing-length convection uses the Ledoux criterion, including the stabilizing effect of composition gradients. Slow mixing associated with semiconvection and thermohaline instability is included. Conduction uses approximate mixtures of Ioffe/Cassisi inputs. Microscopic diffusion, settling and a separate semiconvective heat flux are absent.
Atmosphere	TLUSTY208/SYNSPEC54 calculates plane-parallel gas atmospheres in local thermodynamic equilibrium, resolving absorption and scattering over frequency. The interior joins the atmosphere at optical depth 100. The atmospheres include atomic and molecular absorption and collision-induced absorption, but no accepted condensate treatment.
Thermal neutrinos	Plasma-decay losses follow Haft, Raffelt and Weiss [4] and enter the energy equation with analytic derivatives. Other thermal-neutrino processes are not included in this choice.
Numerical evolution	Coupled implicit equations solve structure, burning and mixing. Comparing a full time step with two half steps controls time accuracy. The mass mesh is fixed. Saved models include enough internal information to continue with the same executable, tables and numerical settings.

The interior equation of state, atmosphere chemistry, opacity and conduction come from different physical sources. Their agreement therefore needs checking; accurate derivatives within one component do not prove that all components describe the same thermodynamics. FreeEOS omits potassium from the GS98 mixture, a missing mass fraction of 4.56×10^{-6} . Helium-isotope and atomic partition-function approximations also remain.

4 Why more atmosphere calculations are needed

The expensive atmosphere calculations are reusable. They are performed separately and stored as tables of boundary temperature and pressure. During stellar evolution, Ember interpolates those tables; it does not solve frequency-dependent radiative transfer at every time step. New atmosphere calculations are needed when the star reaches a composition, surface temperature or gravity not adequately covered, or when the physical treatment changes.

The completed 3.40Tyr track uses 96 atmosphere models, extending down to hydrogen fraction 0.2 and up to surface temperature 3400 K. Not every combination in the listed temperature, gravity and abundance ranges has been calculated. Ember requires the surrounding atmosphere models needed for each interpolation and its derivatives; nominal minimum and maximum values alone are insufficient.

Twelve additional atmospheres at hydrogen fraction 0.1 have converged at 3200 and 3400 K. A trial table containing all 108 atmospheres passes checks of interpolation and compatibility with the interior equation of state at those model positions. Differences between atmosphere and interior densities range from -0.01082% to $+1.991\%$. Previously available temperature and pressure values, and their derivatives, are unchanged at 112 sampled positions. Two independent depth comparisons at the high-

gravity 3200 K extremes change the matching values by less than 1.93×10^{-5} fractionally.

An atmosphere calculated independently at $X_{\text{H}} = 0.15$, $X_3 = 0.005$, 3300 K and $\log g = 5.1$ gives a more demanding interpolation check:

Comparison with the independent atmosphere	Temperature	Gas pressure
Interpolation in all coordinates	+2.134%	−0.7868%
Composition interpolation at fixed temperature and gravity	+1.652%	−0.7842%
Temperature/gravity contribution at surrounding compositions	+0.4743%	−0.002543%

Four extra atmosphere calculations separate the composition contribution from the temperature/gravity contribution. The fractional differences combine multiplicatively, reproducing the full discrepancy. Hydrogen spacing accounts for most of the temperature error. Additional atmospheres at hydrogen fraction 0.15 are therefore being calculated. New independent atmospheres at 0.125 and 0.175 differ from the trial table by 1.539% and 1.833% in matching temperature. A stellar trial using the 108-atmosphere table began losing full convection near 3.543Tyr, then stopped near 3.544Tyr after exceeding the program’s allowed total number of rejected time steps. Its last accepted model had a convective mass fraction of 0.7638. The transition needs further numerical checks and the atmosphere refinement remains unfinished, so this trial is not included in the plotted Ember reference.

Interior composition tables have undergone similar checks. The first zero-hydrogen extension differed from a direct FreeEOS calculation by 2.367% in heat capacity for cool, trace-hydrogen molecular gas. Adding composition points reduced the largest pressure, energy and heat-capacity discrepancies in 528 new comparisons to 0.05213%, 0.1088%, 0.1841% and 0.1922%, respectively. Refining TOPS composition spacing reduced an initial 11.90% opacity discrepancy to a largest difference of 1.706% in the latest specified set of independent comparisons. These measure interpolation differences for the tested states, not the overall physical uncertainty in the sources.

4.1 Condensates and a hydrogen-free atmosphere

Grain formation changes both opacity and thermodynamics. Earlier cool atmosphere calculations failed an enthalpy check, so the condensate treatment has not been accepted for evolution. A new calculation compares gas-only and gas-plus-grain chemical equilibria, including the redistribution of atoms among molecules and ions. At one test state the resulting heat-capacity correction is $1.175 \times 10^5 \text{ erg g}^{-1} \text{ K}^{-1}$, compared with 2.890×10^5 from counting grains alone. The correction still needs a consistent normalization, thermodynamic reference and treatment across phase boundaries before it can be used.

The warmer atmosphere opacity calculations currently reach about 1.010×10^4 K in their deepest layers. A trial extension to 1.500×10^4 K failed in the native helium-line broadening calculation. Reducing the bottom column is acceptable only when the atmosphere remains within the opacity range and its interior matching values no longer depend appreciably on the chosen depth.

A hydrogen-free atmosphere presents a further issue. The source programs’ abundance normalization and molecular initialization currently depend on hydrogen. A direct opacity test at 4017 K shows that multiplying all number abundances by the same factor preserves composition and nearly preserves electron density, but can change individual absorption coefficients by a factor of 27.3. This one-temperature test does not measure a stellar-boundary error; it shows that a simple abundance rescaling is not yet a verified solution. A tiny nonzero hydrogen abundance cannot be labelled zero to evade the problem.

5 Energy and the cold helium remnant

The discrete energy equation used in the reference is

$$L_\gamma = L_{\text{nuc,dep}} + L_{\text{grav}} - L_{\nu,\text{thermal}}, \quad L_{\text{nuc,rest}} = L_{\text{nuc,dep}} + L_{\nu,\text{nuclear}}.$$

Here L_γ is the photon luminosity, $L_{\text{nuc,dep}}$ is nuclear heat left in the star, and L_{grav} is the contribution from changing internal and gravitational energy. Nuclear and thermal neutrinos are counted separately. At 3.40Tyr, in solar luminosity units,

$$L_{\text{nuc,dep}} = 0.002015, \quad L_{\text{grav}} = 6.548 \times 10^{-8}, \quad L_{\nu,\text{nuclear}} = 4.089 \times 10^{-5}, \quad L_{\nu,\text{plasma}} = 3.986 \times 10^{-11}.$$

The changing-energy term refers to the last implicit half step; the other reported diagnostics use the saved state. Before rounding, the discrete luminosity residual is below 10^{-15} . This checks the equations implemented in the code, not an independent exact energy integral or the absence of missing physical processes.

Cold degenerate electrons have a small thermal response superimposed on a large temperature-independent contribution. Subtracting the large terms had lost numerical accuracy. Paired numerical integration around the Fermi surface now retains the small response. Pure-helium tests at 100–10⁴ K and densities 10³–10⁶ g cm⁻³ agree with the expected low-temperature limit to about 10⁻¹⁰ fractionally at 100 K. Electron entropy and ionic translational entropy, including mixing, are also implemented and checked against thermodynamic identities. These analytic components have not yet been combined into the production equation of state for a remnant.

An independent Ioffe EOSFI22 calculation compares liquid and solid free energies and their derivatives. Two quantum pressure-derivative errors were found and repaired in a separate copy of that source. Across sixteen forced-phase tests, the largest checked density-response discrepancy fell from 6.77 to 1.05×10^{-6} in ideal-ion pressure units after refining numerical differentiation. That establishes consistency of the tested derivatives, not which phase is stable or the physical accuracy of the formulas.

Helium quantum melting, mixture effects, latent heat and the connection to partially ionized outer layers remain unfinished. The source’s default classical melting value, $\Gamma = 175$, cannot stand in for a helium phase calculation [5]. Dense, cool atmospheres also require nonideal chemistry and pressure-dependent absorption, including collision-induced absorption; refraction and dense-fluid effects need checking [6].

6 What is required to finish the calculation?

First, the atmosphere and interior inputs must cover the star’s actual path toward hydrogen exhaustion. New calculations must resolve the measured interpolation discrepancies and the hydrogen-free and grain-formation issues, without extrapolating beyond available physical data.

As composition becomes stratified, conservative microscopic diffusion and settling must include the electric field that maintains charge balance and the associated energy exchange. Moving convection boundaries and thin burning layers need enough mass resolution; an adjustable mesh is required if the fixed mesh does not converge. The importance of omitted nuclear reactions, thermal-neutrino processes and semiconvective heat transport must be assessed in the conditions actually reached.

Hydrogen exhaustion and cooling require distinct measurements. We will record central hydrogen fractions crossing 10⁻³, 10⁻⁴ and 10⁻⁵, retaining the accepted models on either side. We will also record when nuclear heat falls below 10%, 1% and 0.1% of photon luminosity during cooling, including

any later recovery. Remaining hydrogen mass and shell burning must be followed; a depleted center alone does not establish that burning has ended everywhere.

Finally, a consistent liquid/solid helium equation of state, conductive transport and a dense atmosphere must allow the star to evolve through 1000, 500 and 100 K on the descending-temperature branch. These endpoints must converge as mass resolution, time accuracy and physical tables are refined. A fitted cooling law or an extrapolated atmosphere is not the requested evolutionary calculation. Numerical error, interpolation error and uncertainty in the physical prescriptions should be reported separately.

Future mass and composition surveys will state the initial helium–metallicity relation and metal pattern explicitly. Burning hydrogen into helium at fixed metal abundance is not equivalent to forming a star at higher metallicity. External heating, encounters and hypothetical particle-physics effects belong to separately defined scenarios [2].

7 Computing time and reproducibility

The Ember calculation to 3.30Tyr used 2351 CPU seconds and 4088 seconds of elapsed time while other source calculations were running. Continuing it to 3.40Tyr used 97.00 CPU seconds. Two separate timing experiments found that reusing exactly identical nuclear calculations reduced CPU time by 12.52% and 10.84%, with identical complete output files. The combined improvement is about 22% for those experiments, not a measured whole-lifetime speedup.

The unmodified F77 AJR run used 27.37 CPU seconds through its programmed cooling stop. Ember’s finer mass mesh, coupled implicit solves, time-step checks and different physics perform additional work. Their separate contributions to the full speed difference have not been measured. An earlier atmosphere replay spent about 54% of its measured time in chemistry and only 0.0033 seconds in the dense matrix solve; that describes one atmosphere calculation, not the stellar integrator.

The [GitHub repository](#) contains code, table-generation instructions, small records of input choices and checks, this paper and the numerical data needed for both figures. The F77 figure data are extracted from accepted models and checked against the recorded output hashes. Full raw outputs, generated physical tables, saved stellar states and executable archives remain local. Instructions for recreating the figures and recovering the recorded stellar calculation accompany the [paper files](#).

The separate build used for this report passes all 32 test suites with the locally installed physical tables. Additional atmosphere archive checks pass after accommodating two added depth descriptions while still requiring exact agreement for every previously recorded value. A fresh checkout needs the locally generated inputs for table-dependent tests and stellar calculations. Detailed input and test records retain full numerical precision and exact file identifiers so the calculations can be checked and reproduced.

References

- [1] Laughlin, G., Bodenheimer, P., & Adams, F. C. (1997). The End of the Main Sequence. *ApJ*, 482, 420. [doi:10.1086/304125](#).
- [2] Adams, F. C., & Laughlin, G. (1997). A dying universe: the long-term fate and evolution of astrophysical objects. *Rev. Mod. Phys.*, 69, 337. [doi:10.1103/RevModPhys.69.337](#).
- [3] Baraffe, I., Chabrier, G., Allard, F., & Hauschildt, P. H. (1998). Evolutionary models for solar metallicity low-mass stars: mass–magnitude relationships and color–magnitude diagrams. *A&A*, 337, 403. [arXiv:astro-ph/9805009](#).

- [4] Haft, M., Raffelt, G., & Weiss, A. (1994). Standard and nonstandard plasma neutrino emission revisited. [arXiv:astro-ph/9309014](#).
- [5] Baiko, D. A., & Chugunov, A. I. (2022). *Ab initio* thermodynamics of one-component plasma for astrophysics of white dwarfs and neutron stars. [arXiv:2112.04822](#).
- [6] Blouin, S., Dufour, P., & Allard, N. F. (2018). A new generation of cool white dwarf atmosphere models. I. Theoretical framework and applications to DZ stars. [arXiv:1807.06616](#).